Vectors I: Basics Worksheet

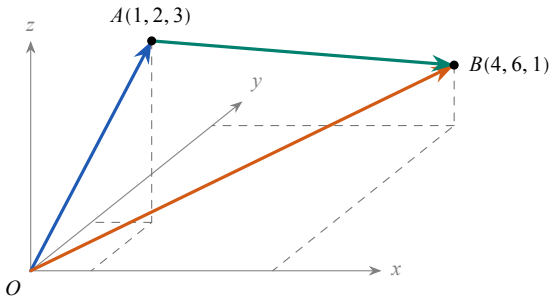
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Basics of vectors: **i**, **j**, **k** notation, magnitude, unit vectors, parallel vectors, collinear points, the ratio theorem, the scalar product and the vector product. Total: 29 marks (Practice Problems; sections 1–8 are unmarked exercises).

1 **i**, **j**, **k** Notation and Column Vectors

1 Write $\mathbf{a}=3\mathbf{i}-2\mathbf{j}+5\mathbf{k}$ as a column vector.

2 The points A and B have coordinates $(1, 2, 3)$ and $(4, 6, 1)$. Find $\overrightarrow{AB}$.



3 Given $\mathbf{p}=2\mathbf{i}+\mathbf{k}$ and $\mathbf{q}=-\mathbf{i}+3\mathbf{j}$, find $\mathbf{r}$ such that $2\mathbf{p}-3\mathbf{q}+\mathbf{r}=\mathbf{0}$.

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{i}+\mathbf{j}+\mathbf{k}=\begin{pmatrix}1\\1\\1\end{pmatrix}$	
(b)	The **j**-component of $4\mathbf{i}-\mathbf{j}+2\mathbf{k}$ is -1	
(c)	The position vector of $P(1, -2, 0)$ is $\mathbf{i}-2\mathbf{j}$	
(d)	$3\mathbf{i}+2\mathbf{k}=\begin{pmatrix}3\\2\\0\end{pmatrix}$	
(e)	$\overrightarrow{AB}=\overrightarrow{OA}-\overrightarrow{OB}$	

2 Magnitude

1 Find $\left|\begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}\right|$.

2 The points A and B have coordinates $(1, 2, -1)$ and $(3, -4, 2)$. Find the distance AB.

3 Given $\mathbf{a} = \begin{pmatrix} t \\ 2 \\ t-1 \end{pmatrix}$ and $|\mathbf{a}| = 3$, find the possible values of t.

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\lvert-\mathbf{a}\rvert = \lvert\mathbf{a}\rvert$ for every vector $\mathbf{a}$	
(b)	$\lvert 3\mathbf{a}\rvert = 3\lvert\mathbf{a}\rvert$	
(c)	$\left\lvert\begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix}\right\rvert = -5$	
(d)	$\left\lvert\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\rvert = 1$	
(e)	$\left\lvert\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}\right\rvert = 1+2-2=1$	
(f)	$\lvert\mathbf{a}+\mathbf{b}\rvert = \lvert\mathbf{a}\rvert + \lvert\mathbf{b}\rvert$ for all vectors $\mathbf{a}$, $\mathbf{b}$	
(g)	$\lvert\mathbf{a}+\mathbf{b}\rvert \leq \lvert\mathbf{a}\rvert + \lvert\mathbf{b}\rvert$ for all vectors $\mathbf{a}$, $\mathbf{b}$	
(h)	For all $\lambda \in \mathbb{R}$, $\lvert\lambda\mathbf{a}\rvert = \lambda\lvert\mathbf{a}\rvert$	

3 Unit Vectors

1 Find the unit vector in the direction of $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

2 Find the vector of magnitude 10 in the direction of $\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$.

3 The points A and B have coordinates $(1, 0, 2)$ and $(3, -6, 5)$. Find the vector of magnitude 14 in the direction *opposite* to $\overrightarrow{AB}$.

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is a unit vector	
(b)	$\frac{1}{3}\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ is a unit vector	
(c)	$\mathbf{i}+\mathbf{j}$ is a unit vector	
(d)	$\lvert\hat{\mathbf{a}}\rvert = 1$ for every non-zero vector $\mathbf{a}$	
(e)	$\mathbf{a}$ and $\hat{\mathbf{a}}$ are parallel	

4 Parallel Vectors ($\mathbf{a} = \lambda \mathbf{b}$)

1 Show that $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -6 \\ 3 \\ -9 \end{pmatrix}$ are parallel.

2 Find p and q such that $\begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix}$.

3 Find all vectors of magnitude 6 that are parallel to $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$.

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	If $\mathbf{a} \parallel \mathbf{b}$, then $\mathbf{a} = \mathbf{b}$	
(b)	$\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$	
(c)	$\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$	
(d)	$\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$	
(e)	If $\mathbf{a} = -2\mathbf{b}$, then $\mathbf{a}$ and $\mathbf{b}$ are not parallel since they point in opposite directions	

5 Collinear Points

1 Show that the points $A(1, 2, 3)$, $B(3, 5, 4)$ and $C(7, 11, 6)$ are collinear.

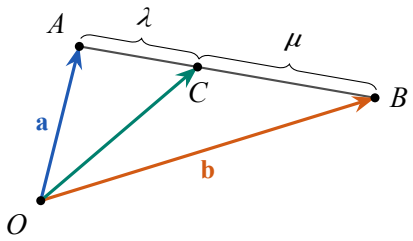
2 For the points in Question 1, find the ratio $AB : BC$.

3 The points $P(2, -1, 0)$, $Q(4, 1, 3)$ and $R(a, 7, b)$ are collinear. Find a and b.

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	If $\overrightarrow{AB} \parallel \overrightarrow{CD}$, then A, B, C and D are collinear	
(b)	If $\overrightarrow{AB} = 3\overrightarrow{AC}$, then A, B and C are collinear	
(c)	$A(0, 0, 0)$, $B(1, 1, 1)$ and $C(2, 2, 3)$ are collinear	
(d)	The points with position vectors $\mathbf{a}$, $2\mathbf{a}$ and $-\mathbf{a}$ are collinear ($\mathbf{a} \neq \mathbf{0}$)	

6 Ratio Theorem



In the True/False statements, C lies on AB with $AC:CB=\lambda:\mu$, $\overrightarrow{OA}=\mathbf{a}$ and $\overrightarrow{OB}=\mathbf{b}$.

1 Let $\overrightarrow{OA}=\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\overrightarrow{OB}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$. Given that C lies on AB, find the coordinates of C if

(a) $AC:CB=1:2$,

(b) $3AC=2CB$,

(c) $2AB=5CB$,

(d) Given instead that C lies on AB produced, and $3AC=7BC$.

2 Find the position vector of the midpoint M of AB, where A and B have coordinates $(3,-1,5)$ and $(-1,5,1)$.

3 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\overrightarrow{OC}=\dfrac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$	
(b)	$\overrightarrow{OC}=\dfrac{\lambda\mathbf{a}+\mu\mathbf{b}}{\lambda+\mu}$	
(c)	If $\lambda:\mu=1:1$, then $\overrightarrow{OC}=\dfrac{1}{2}(\mathbf{a}+\mathbf{b})$	
(d)	If $\lambda:\mu=1:3$, then $\overrightarrow{OC}=\dfrac{3\mathbf{a}+\mathbf{b}}{4}$	
(e)	If $\lambda:\mu=2:1$, then $\overrightarrow{OC}=\dfrac{2\mathbf{a}+\mathbf{b}}{3}$	

7 Scalar Product

1 Properties of scalar product

It is given that $\mathbf{a}=\begin{pmatrix}1\\2\\3\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-3\\2\\7\end{pmatrix}$, $\mathbf{c}=\begin{pmatrix}4\\-5\\-1\end{pmatrix}$. Find the following

(a) $\mathbf{a}\cdot\mathbf{b}$ (b) $\mathbf{b}\cdot\mathbf{a}$ (c) $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})$ (d) $\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$

(e) $\mathbf{a}\cdot(2\mathbf{c})$ (f) $(2\mathbf{a})\cdot\mathbf{c}$ (g) $\mathbf{a}\cdot\mathbf{a}$

2 Scalar product formula: $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta$

(a) Points O, A, B are such that $\overrightarrow{OA}=\mathbf{i}+4\mathbf{j}-3\mathbf{k}$ and $\overrightarrow{OB}=5\mathbf{i}-\mathbf{j}$

(i) Find the size of angle AOB

(ii) Find the size of angle OAB.

(b) It is given that $\mathbf{c}=\begin{pmatrix}2\\3\\6\end{pmatrix}$, $\mathbf{d}=\begin{pmatrix}1\\-2\\3\end{pmatrix}$ and $\mathbf{e}=\begin{pmatrix}3\\2\\-2\end{pmatrix}$. Find the angle between the vectors $\mathbf{c}$ and

(i) the vector $\mathbf{d}$

(ii) the vector $\mathbf{e}$

(iii) the positive y-axis

(c) It is given that two vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy the following relationship:

$$|\mathbf{a}||\mathbf{b}|=6.$$

Find the value of $\mathbf{a}\cdot\mathbf{b}$ for each of the following scenarios.

(i) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $60°$.

(ii) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $120°$.

(iii) Vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

3 Perpendicularity of vectors

(a) Referred to the origin O, the position vectors of the points A and B are $\mathbf{i}-\mathbf{j}+2\mathbf{k}$ and $2\mathbf{i}+4\mathbf{j}+\mathbf{k}$ respectively. Show that OA is perpendicular to OB.

(b) The position vectors $\mathbf{a}$ and $\mathbf{b}$ are given by

$$\mathbf{a}=2p\mathbf{i}+3p\mathbf{j}+6p\mathbf{k} \quad \text{and} \quad \mathbf{b}=\mathbf{i}-2\mathbf{j}+2\mathbf{k}$$

where $p>0$. It is given that $|\mathbf{a}|=|\mathbf{b}|$.

(i) Find the exact value of p.

(ii) Show that the vectors $(\mathbf{a}+\mathbf{b})$ and $(\mathbf{a}-\mathbf{b})$ are perpendicular.

4 Length of projection

(a) Given that $\mathbf{a}=2\mathbf{i}+\mathbf{j}+2\mathbf{k}$ and $\mathbf{b}=-\mathbf{i}+2\mathbf{j}-\mathbf{k}$, find the length of projection and the projection vector of

(i) $\mathbf{a}$ onto $\mathbf{b}$,

(ii) $\mathbf{b}$ onto $\mathbf{a}$.

(b) With reference to the origin O, the points A, B and C have position vectors given by

$$\mathbf{a}=\mathbf{i}+2\mathbf{j}-\mathbf{k},\ \mathbf{b}=3\mathbf{i}+\mathbf{k} \text{ and } \mathbf{c}=2\mathbf{i}+2\mathbf{j}+3\mathbf{k}$$

respectively. Find the length of projection of $\overrightarrow{BA}$ onto $\overrightarrow{BC}$.

8 Vector Product

1 Properties of vector product

(a) Find $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix}$

(b) Given that $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$, find

(i) $\mathbf{a} \times \mathbf{b}$; (ii) $\mathbf{b} \times \mathbf{a}$; (iii) $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$; (iv) $\mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$;

(v) $\mathbf{a} \times (2\mathbf{b})$; (vi) $(2\mathbf{a}) \times \mathbf{b}$; (vii) $\mathbf{a} \times \mathbf{a}$;

2 Parallel vectors

(a) Three non-zero vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy $\mathbf{a} \times 3\mathbf{b} = 2\mathbf{a} \times \mathbf{c}$. Show that $\mathbf{a}$ and $3\mathbf{b} - 2\mathbf{c}$ are parallel.

(b) Find the scalar product and vector product of the following vectors, stating whether they are parallel, perpendicular, or neither.

(i) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$

(ii) $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + \mathbf{k}$

(iii) $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ -8 \\ 4 \end{pmatrix}$

3 Vector product formula

Using the Vector Product, find the acute angles between the following vectors:

(a) $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$ (b) $\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$ (c) $\begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$

4 Perpendicular distance

(a) Relative to an origin O, two points A and B have position vectors given by $\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = -2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$. Find the following:

(i) The shortest distance from A to OB

(ii) The shortest distance from B to OA

(b) Relative to the origin O, two points A and B have position vectors given by $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ respectively.

(i) Find the length of projection of $\mathbf{b}$ onto $\mathbf{a}$.

(ii) Hence, or otherwise, find the shortest distance from B to OA.

(c) Relative to the origin O, the points A, B and C have position vectors given by $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{c} = 3\mathbf{i} - \mathbf{j} + \mathbf{k}$. Find the shortest distance from C to the line joining A and B.

5 Areas of triangles & parallelograms

(a) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix}$. Find the area of triangle OAB.

(b) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$. Find the area of triangle OAB.

(c) Relative to an origin O, the position vectors of the vertices A, B and C of a triangle are

$$\mathbf{i}+\mathbf{j}-\mathbf{k},\ 2\mathbf{i}+\mathbf{k} \text{ and } -\mathbf{i}+2\mathbf{j}-\mathbf{k}$$

respectively. By considering the displacement vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$, find the exact area of triangle ABC.

9 Practice Problems

1 Relative to the origin O, the position vectors of two points A and B are $\mathbf{a}$ and $\mathbf{b}$ respectively. The length of $\mathbf{a}$ is 4 units and $\mathbf{b}$ is a unit vector. The angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{3}$ radians.

(a) Find the area of triangle OAB. [2]

It is given that the point M lies on the line segment AB such that $AM : AB$ is $3 : 4$.

(b) Find the exact area of triangle OAM. [3]

It is given that the point C lies on AB produced, such that $\overrightarrow{AC} = \lambda \overrightarrow{AB}$, where $\lambda > 1$.

(c) Find the value of λ such that OC is perpendicular to AB. [4]

2 The non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy $|\mathbf{a}| = 3$, $|\mathbf{b}| = 4$ and $\mathbf{a} \cdot \mathbf{b} = 6$.

(a) Find the angle between $\mathbf{a}$ and $\mathbf{b}$. [2]

(b) Find $|\mathbf{a} - \mathbf{b}|$. [3]

3 Relative to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $3\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel.

(a) Find the area of triangle OAC in terms of $|\mathbf{a} \times \mathbf{b}|$. [2]

(b) The point D is the midpoint of AC. Find the area of triangle OAD in terms of $|\mathbf{a} \times \mathbf{b}|$. [2]

4 The non-zero vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy $3\mathbf{a} \times \mathbf{b} = 2\mathbf{c} \times \mathbf{b}$.

(a) Show that $3\mathbf{a} - 2\mathbf{c} = \lambda \mathbf{b}$ for some scalar λ. [2]

(b) It is now given that $|\mathbf{a}| = 2$, $|\mathbf{b}| = 3$ and the angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{6}$. Find $|3\mathbf{a} - 2\mathbf{b}|$. [3]

5 Relative to the origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. It is given that $|\mathbf{a}| = 2$ and that $\mathbf{b}$ is a unit vector. The point C divides the line segment AB internally such that $AC : CB = 3 : 2$. The point D lies on OB produced such that $\overrightarrow{OD} = 3\overrightarrow{OB}$.

(a) Find $\overrightarrow{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(b) Find $\overrightarrow{OD}$ in terms of $\mathbf{b}$. [1]

(c) Given that CD is perpendicular to AB, find the value of $\mathbf{a} \cdot \mathbf{b}$. [3]

END OF WORKSHEET